

Def.

A space  $X$  is called a Kolmogorov (or  $T_0$ ) space if: given  $x, y \in X$  with  $x \neq y$ , there exists an open set  $U \subseteq X$  such that either  $\begin{cases} x \in U \\ y \notin U \end{cases}$  or  $\begin{cases} x \notin U \\ y \in U \end{cases}$ .

Prop.

$X$  is  $T_0$  space if and only if  $x \neq y \Rightarrow \overline{\{x\}} \neq \overline{\{y\}}$

Def.

A space  $X$  is called a Fréchet space (or  $T_1$ ) if: given  $x, y \in X$  with  $x \neq y$ , there exists open sets  $U, V \subseteq X$  such that  $\begin{cases} x \in U \\ y \notin U \end{cases}$  and  $\begin{cases} x \notin V \\ y \in V \end{cases}$

Prop.

$X$  is  $T_1$ -space if and only if for all  $x \in X$ ,  $\overline{\{x\}} = \{x\}$ .

Proof. Suppose that  $X$  is  $T_1$ -space.